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Project Northstar – BA Portfolio

This space contains the Project Northstar Business Analysis portfolio. Start with page 00 – ProjectOverview.

00 – Project Overview (Start Here)

Project Northstar: Asset & Work Order System Replacement

A self-directed Business Analysis portfolio project by **Keerthi Ajantha Kuchipudi**

About this project

This portfolio project simulates the Business Analyst role on a council systems replacement. Banyan Coast Council (fictional) is replacing a legacy Access database and spreadsheets with a cloud-based asset and work order management system, integrated with finance and GIS. The council, stakeholders and workshop are fictional. The data analysis uses a real open dataset: the City of Melbourne "Street lights with emitted lux level" dataset (106,037 records, CC BY licence).

The problem

Maintenance requests were lost or duplicated, crews were sent to wrong locations, finance staff re-keyed repair costs by hand, and monthly reports took around five days to compile.

What I did

- Defined scope, objectives and stakeholders in a project brief
- Captured requirements, decisions and actions from a simulated stakeholder workshop
- Tracked risks, assumptions, issues and dependencies in a RAID log
- Mapped the current and future state processes using swimlane diagrams, and completed a gap analysis
- Wrote 23 requirements as user stories with acceptance criteria, prioritised using MoSCoW
- Managed the requirements in a Jira Scrum backlog, grouped into 6 epics
- Profiled 106,037 legacy records in Excel and built a source-to-target data mapping
- Tested the mapping rules on a sample of real records
- Specified the finance integration (fields, business rules, error handling) and reporting requirements
- Wrote UAT test scenarios, a defect log and a requirements traceability matrix

Key findings from the data analysis

- Only one field (ext_id) was a reliable unique identifier: 100% populated, 0 duplicates.
- Location data existed only in one combined field; easting and northing were 100% zero.
- 8,632 records shared coordinates with another record, needing review before migration.
- 662 lights recorded 0 lux, flagged for inspection.
- The source data had no suburb field, which created a dependency for suburb-based reporting. This was raised in the RAID log and traced through to a blocked test.

Tools used

Confluence, Jira, Confluence Whiteboards, Microsoft Excel (COUNTIF, COUNTBLANK, UNIQUE, TEXTBEFORE/TEXTAFTER, DATE)

Project pages

Page	What it shows
01 – Project Brief	Background, problem, objectives, scope, stakeholders
02 – Stakeholders & Workshop Notes	Stakeholder analysis, workshop notes, decisions and actions
03 – RAID Log	Risks, assumptions, issues and dependencies
04 – Current State Process Map	How fault requests are handled today, with pain points
05 – Future State Process Map	How the process works in the new system
06 – Gap Analysis	What must change, linked to requirements
07 – Requirements Register	23 requirements with acceptance criteria and MoSCoW priority
08 – Data Profiling & Mapping	Data quality findings, mapping and transformation test (Excel attached)
09 – Integration & Reporting Requirements	Finance interface specification and report definitions
10 – Test Scenarios & Traceability	UAT scenarios, defects and traceability matrix

How the work connects

Workshop pain point → process map → gap → requirement → data mapping → test case. For example: crews going to the wrong location (workshop) → P2 on the current state map → gap G-02 → REQ-009 and REQ-011 → coordinates split in the data mapping → TC-10 executed in Excel.

01 – Project Brief

Project Northstar

Asset & Work Order Management System Replacement

Organisation: Banyan Coast Council (fictional), Northern Territory

Prepared by: Keerthi Ajantha Kuchipudi, Business Analyst

Version: 0.1 (Draft) | **Date:** 24 Sep 2026

1. **Background**

   Banyan Coast Council manages over 8,000 community assets, including streetlights, roads, parks, public toilets and council buildings. Residents report faults by phone, email or at the front counter. Currently, assets and maintenance requests are tracked in a legacy Microsoft Access database and several Excel spreadsheets maintained by different teams.

2. **Problem Statement**

   The current approach causes several issues:

- Maintenance requests are lost or duplicated because there is no single system.
- Managers cannot report on open jobs, response times or maintenance costs.
- Finance staff manually re-enter repair costs into the finance system, causing delays and errors.
- Asset data is inconsistent, with duplicate records and missing information.

3. **Objectives**

- Replace the legacy database and spreadsheets with a single cloud-based asset and work order management system.
- Transfer completed work order costs to the council's finance system through a scheduled nightly batch integration, removing manual re-entry (integration method to be confirmed with the vendor and Finance).
- Display asset locations in the council's GIS by consuming its existing spatial data layers.
- Provide managers with dashboard reporting on open work orders, response times and costs, reducing reporting lead time from approximately 5 working days to same-day.
- Migrate clean, accurate asset data from the legacy system, with zero duplicate asset IDs at go-live.

4. **Scope**

   In scope: asset register, work order management, finance system integration, GIS integration, reporting, and data migration from the legacy Access database and spreadsheets.

   Out of scope: payroll, a mobile app for field crews (planned for Phase 2), and replacement of the finance or GIS systems.

5. **Key Stakeholders**

Stakeholder	Role	Main interest
Works Manager	Project sponsor	Visibility of jobs and costs
Field Crew Supervisor	System user	Clear job assignments
Customer Service Officer	System user	Fast, easy request logging
Finance Officer	Integration user	Accurate cost data, no re-keying
GIS Officer	Technical stakeholder	Asset data compatible with existing spatial layers and coordinate standards
Software Vendor	Solution provider	Clear requirements and configuration

6. **Assumptions & Constraints**

- The new system has already been selected; this project covers requirements, configuration support, data migration and testing.
- Legacy data quality is unknown and must be assessed before migration.
- Go-live is planned within 9 months.
- Baseline figures (such as the 5-day reporting lead time) are initial estimates and will be validated with stakeholders during workshops.

7. **BA Deliverables**
   Workshop notes and RAID log, current and future state process maps, requirements register, data profiling and mapping, interface and reporting requirements, test scenarios, and a requirements traceability matrix.

02 – Stakeholders & Workshop Notes

Note: This is a simulated workshop created for portfolio purposes. Stakeholders and discussions are fictional, based on common challenges in council asset management.

Project: Project Northstar | **Version:** 0.1 (Draft) | **Author:** Keerthi Ajantha Kuchipudi

1. **Stakeholder Analysis**

Stakeholder	Name (fictional)	Influence	Interest	Engagement approach
Works Manager (Sponsor)	Daniel Reid	High	High	Manage closely: fortnightly updates, approves requirements
Project Manager	Lisa Moreno	High	High	Manage closely: daily coordination
Field Crew Supervisor	Tom Walsh	Medium	High	Keep informed: involve in process and testing workshops
Customer Service Officer	Priya Nair	Low	High	Keep informed: involve in request logging design
Finance Officer	Sarah Chen	Medium	High	Manage closely on integration requirements

GIS Officer	Ben Okafor	Medium	Medium	Consult on spatial data and asset locations
Vendor Implementation Lead	Mark Ellis (AssetCloud Pty Ltd)	High	High	Weekly working sessions on configuration and integration

2. **Workshop 1: Current State & Requirements Discovery**

Date: 24 Sept 2026 | **Duration:** 2 hours | **Facilitator:** Lisa Moreno | **Notes:** Keerthi Ajantha Kuchipudi (BA)
Attendees: Daniel Reid, Tom Walsh, Priya Nair, Sarah Chen, Ben Okafor, Mark Ellis

Purpose: Understand how maintenance requests are handled today, identify key pain points, and capture initial requirements for the new system.

Key discussion points

- **Customer Service (Priya):** Requests are logged in an Excel sheet and emailed to supervisors. There is no way to check the status of a request when a resident calls back.
- **Field Crews (Tom):** Supervisors receive requests by email and assign jobs using a whiteboard. Crews sometimes arrive at the wrong location because asset details are unclear.
- **Finance (Sarah):** Repair costs are submitted on paper timesheets and re-entered manually. This takes around 2 days per month and causes errors.
- **GIS (Ben):** Asset locations in the Access database do not match GIS coordinates for many records. The new system must use the council's existing spatial layers.
- **Works Manager (Daniel):** Monthly reports take around 5 working days to compile manually. He needs a dashboard showing open jobs and response times.
- **Vendor (Mark):** The new system supports both API and file-based (batch) integration with finance systems.

Initial requirements identified *(to be added to the Requirements Register)*

- Customer service must be able to log a request against a specific asset and give residents a reference number.

- Supervisors must be able to assign work orders to crews in the system.
- Staff must be able to check the status of any request.
- Completed work order costs must transfer to the finance system without manual re-entry.
- Asset locations must display on a map using existing GIS layers.
- Managers need a dashboard of open work orders and response times.

Decisions

ID	Decision	Made by
D-01	Mobile app for field crews is out of scope for Phase 1	Daniel Reid
D-02	Nightly batch is the preferred finance integration method, subject to vendor confirmation	Sarah Chen, Mark Ellis

Actions

ID	Action	Owner	Due date	Status
A-01	Provide a sample export of the legacy Access asset data	Tom Walsh	1 Oct 2026	Open
A-02	Confirm the finance system's file format for cost imports	Sarah Chen	3 Oct 2026	Open
A-03	Share GIS spatial layer	Ben Okafor	3 Oct 2026	Open

	specifications			
A-04	Confirm batch integration capability and file structure	Mark Ellis	3 Oct 2026	Open
A-05	Draft current and future state process maps	Keerthi (BA)	8 Oct 2026	Open

Open questions

- Who approves work orders over a certain cost threshold, and what is that threshold?
- Do residents need automatic notifications (SMS/email) when a request is completed?
- How many years of historical work order data should be migrated?

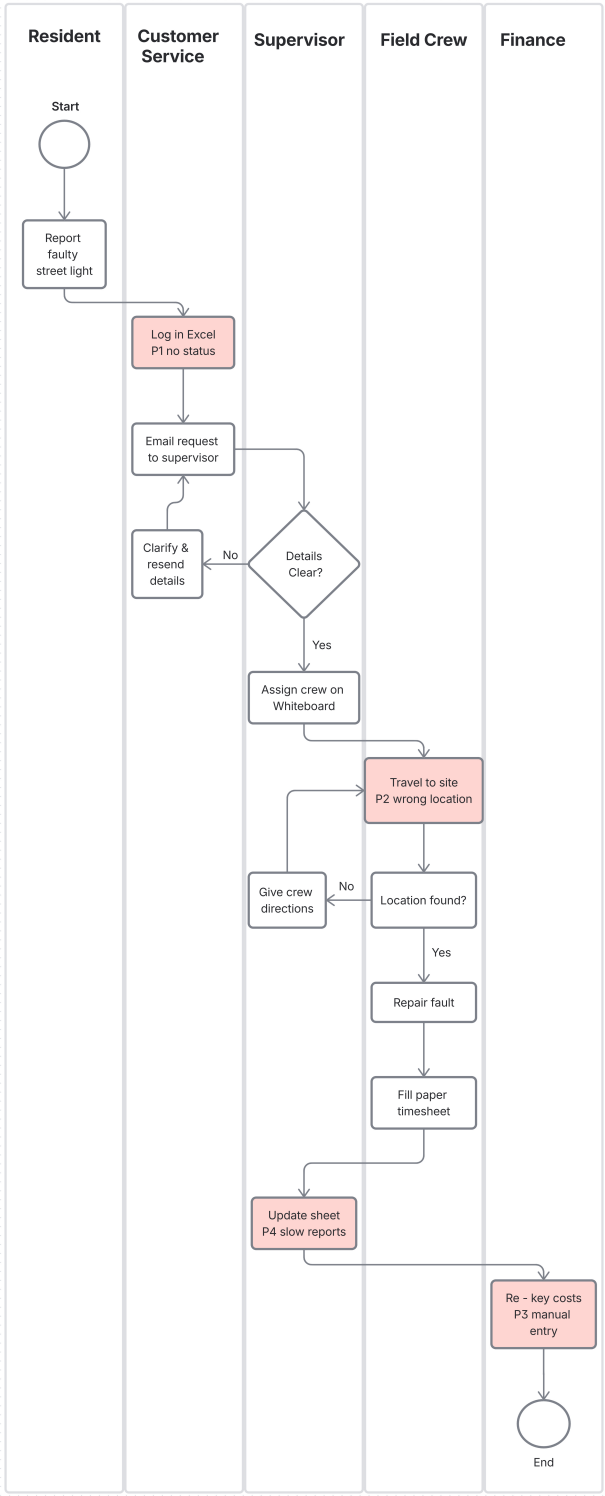
03 – RAID Log

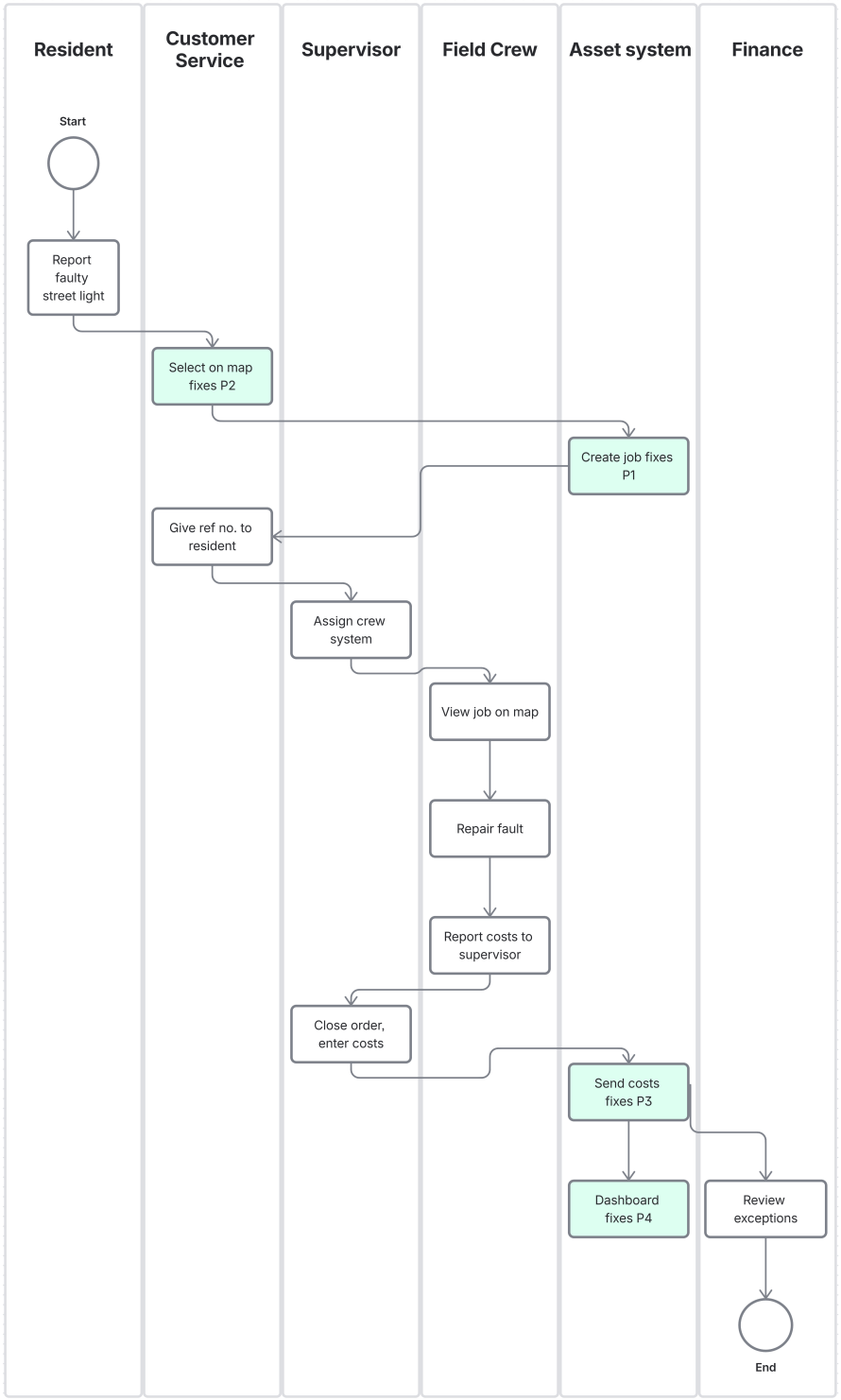
Project: Project Northstar | **Last updated:** 9 October 2026 | **Owner:** Keerthi Ajantha Kuchipudi

Purpose: This log tracks the project's Risks, Assumptions, Issues and Dependencies (RAID). It is reviewed weekly with the Project Manager.

ID	Type	Description	Impact	Owner	Response / Mitigation	Status
R-01	Risk	Legacy asset data contains duplicates and missing values	Migration failures or incorrect data at go-live	BA	Profile data early and agree cleansing rules with asset owners	Open – confirmed by data profiling (see page 08)
R-02	Risk	Asset coordinates in Access do not match GIS layers	Crews sent to wrong locations	Ben Okafor	Validate coordinates against GIS layers before migration	Open
R-03	Risk	Field staff may resist moving away from whiteboard and email	Low adoption of new system	Daniel Reid	Involve crews in testing and provide training before go-live	Open
A-01	Assumption	Finance system can accept a nightly cost import file	Integration design depends on this	Sarah Chen	Confirm via workshop action A-02	Open
A-02	Assumption	5-day reporting lead time is accurate	Success metric may be wrong	BA	Validate with Works Manager	Open

I-01	Issue	No single owner for asset data across teams	Delays in data cleansing decisions	Daniel Reid	Nominate a data owner for each asset type	Open
D-01	Dependency	Vendor confirmation of batch integration capability	Finance integration design cannot be finalised	Mark Ellis	Tracked via workshop action A-04	Open
D-02	Dependency	Access to legacy data export	Data profiling and mapping cannot start	Tom Walsh	Tracked via workshop action A-01	Open
R-04	Risk	8,632 records share coordinates with another record	Duplicate assets may be migrated	Ben Okafor	Review before migration (DQ-02)	Open
I-02	Issue	Source data has no suburb field	Suburb reporting (REQ-015) not possible without GIS enrichment	Ben Okafor	Derive suburb from GIS boundaries (DQ-05)	Open
A-03	Assumption	xdate represents the data capture (survey) date	Date field may be mapped incorrectly	Ben Okafor	Confirm meaning (DQ-04)	Open


Resident
Customer Service
Supervisor
Field Crew
Finance
Start
Report faulty street light
Log in Excel P1 no status
Email request to supervisor
Clarify & resend details
No
Details Clear?
Yes
Assign crew on Whiteboard
Travel to site P2 wrong location
Give crew directions
No
Location found?
Yes
Repair fault
Fill paper timesheet
Update sheet P4 slow reports
Re - key costs P3 manual entry
End


Resident
Customer Service
Supervisor
Field Crew
Asset system
Finance
Start
Report faulty street light
Select on map fixes P2
Create job fixes P1
Give ref no. to resident
Assign crew system
View job on map
Repair fault
Report costs to supervisor
Close order, enter costs
Send costs fixes P3
Dashboard fixes P4
Review exceptions
End

06 – Gap Analysis

Project: Project Northstar | **Version:** 0.1 (Draft) | **Author:** Keerthi Ajantha Kuchipudi

Purpose: This page compares the current state and future state process maps for the streetlight fault request process. Each gap describes what must change, and will be traced to requirements in the Requirements Register.

Related pages: 04 – Current State Process Map, 05 – Future State Process Map

Gap ID	Pain point	Current state	Future state	What needs to change
G-01	P1: No request status	Requests logged in Excel; staff cannot see progress	System creates a work order with a reference number; status visible to all staff	System must create and track work orders with a unique reference number and status
G-02	P2: Wrong location	Location described in free text; crews often go to the wrong site	Customer Service selects the asset on a map; crew sees the location	System must link requests to specific assets and display them on a GIS map
G-03	P3: Manual cost entry	Finance re-enters costs from paper timesheets	Costs sent to finance automatically in a nightly batch; Finance reviews exceptions only	System must export completed work order costs to the finance system daily
G-04	P4: Slow reporting	Monthly reports compiled manually over ~5 days	Dashboard updates automatically from system data	System must provide a dashboard of open work orders, response times and costs
G-05	Rework loops	Supervisor and crew call back and forth to clarify	Details and location captured	System must make asset selection

		details and locations	correctly at the start	mandatory when logging a request
G-06	Manual job assignment	Jobs assigned on a whiteboard	Supervisor assigns crews in the system	System must allow supervisors to assign work orders to crews

Out of scope note: Field crews will not use the system directly in Phase 1 (Decision D-01). A mobile app for crews is planned for Phase 2.

07 – Requirements Register

Project: Project Northstar | **Version:** 0.1 (Draft) | **Author:** Keerthi Ajantha Kuchipudi

Purpose: This register lists the requirements for the new asset and work order management system. Each requirement is traced to its source (a gap from the Gap Analysis, a workshop item or a RAID entry) and will be traced to test scenarios.

Priority key (MoSCoW): Must = essential for go-live | Should = important but not critical | Could = desirable if time allows

Related pages: 02 – Stakeholders & Workshop Notes, 03 – RAID Log, 06 – Gap Analysis

Epic 1: Request & Work Order Management

ID	Type	Requirement	Acceptance criteria	Priority	Source
REQ-001	Functional	As a Customer Service Officer, I want to log a request against a specific asset, so that the correct asset is repaired	A request cannot be saved without a valid asset ID	Must	G-02, G-05
REQ-002	Functional	As a Customer Service Officer, I want the system to generate a unique reference number for each request, so that I can give it to the resident	A unique reference number is displayed when the request is saved	Must	G-01

REQ-003	Functional	As a Customer Service Officer, I want to search requests by reference number and see their status, so that I can update residents who call back	Search returns the request and its status (Open, Assigned, Completed)	Must	G-01
REQ-004	Functional	As a Supervisor, I want to assign work orders to a crew in the system, so that all jobs are tracked in one place	Assigned crew and date are recorded; status changes to Assigned	Must	G-06
REQ-005	Functional	As a Supervisor, I want to print a job sheet showing the asset's map location, so that crews can find the correct site	Job sheet includes asset ID, address, coordinates and a map image	Must	G-02, D-01
REQ-006	Functional	As a Supervisor, I want to close a work order and	A work order cannot be closed without at least one	Must	G-03

		record labour and material costs, so that costs are captured accurately	cost line and a cost code		
REQ-007	Functional	As a Works Manager, I want work orders above a cost threshold to require my approval, so that spending is controlled	Work orders above the threshold are held for approval (threshold TBC)	Should	Workshop open question
REQ-008	Functional	As a Resident, I want to receive an SMS or email when my request is completed, so that I know the fault is fixed	Notification sent within 1 hour of work order closure	Could	Workshop open question

Epic 2: Asset Register & GIS

ID	Type	Requirement	Acceptance criteria	Priority	Source
REQ-009	Integration	As a Customer Service Officer, I want to select an asset on a map, so that I can identify it accurately	Map displays assets from the council's existing GIS layers; clicking an asset selects it	Must	G-02

REQ-010	Data	Each asset record must store asset ID, asset type, coordinates, suburb, install date and condition	All listed fields exist; asset ID and asset type are mandatory	Must	Workshop (GIS Officer)
REQ-011	Data	Asset coordinates must match the council's GIS coordinate standard	Migrated asset locations display in the correct position on the GIS map	Must	R-02

Epic 3: Finance Integration

ID	Type	Requirement	Acceptance criteria	Priority	Source
REQ-012	Integration	The system must send completed work order costs to the finance system in a nightly batch	Costs appear in the finance system by 7am the next business day	Must	G-03, Decision D-02
REQ-013	Integration	As a Finance Officer, I want a daily error report of records that failed to import, so that I can correct them	Failed records are listed with the reason for failure	Must	G-03

REQ-014	Integration	The integration must not send the same work order cost more than once	Re-running a batch does not create duplicate cost entries	Must	G-03

Epic 4: Reporting

ID	Type	Requirement	Acceptance criteria	Priority	Source
REQ-015	Reporting	As a Works Manager, I want a dashboard of open work orders by suburb and asset type, so that I can see workload at a glance	Dashboard shows current open work orders, filterable by suburb and asset type	Must	G-04
REQ-016	Reporting	As a Works Manager, I want to see the average response time per month, so that I can monitor service levels	Report shows average days from request logged to work order closed	Should	G-04
REQ-017	Reporting	As a Works Manager, I want to see monthly maintenance	Report shows total costs per asset type per month	Should	G-04

		costs by asset type, so that I can plan budgets			

Epic 5: Data Migration

ID	Type	Requirement	Acceptance criteria	Priority	Source
REQ-018	Data	All active assets from the legacy Access database must be migrated	Migrated asset count matches the agreed source count; zero duplicate asset IDs	Must	R-01, Brief objective
REQ-019	Data	Open maintenance requests at go-live must be migrated with their current status	All open requests appear in the new system with correct status	Must	Workshop
REQ-020	Data	Historical work orders must be migrated for reporting	Agreed number of years of history migrated (TBC)	Could	Workshop open question

Epic 6: Non-Functional

ID	Type	Requirement	Acceptance criteria	Priority	Source
REQ-021	Non-functional	The system must be accessible through a web	Staff can log in using a standard browser	Must	Vendor solution

		browser without local installation			
REQ-022	Security	Access must be role-based	Only Supervisors can close work orders; only Finance can view cost reports	Must	Workshop
REQ-023	Non-functional	The system must be available during business hours	99.5% availability between 7am and 7pm on business days	Should	Brief

08 – Data Profiling & Mapping

Project: Project Northstar | **Version:** 0.1 (Draft) | **Author:** Keerthi Ajantha Kuchipudi

Purpose: This page summarises the profiling of the legacy streetlight asset data, the proposed source-to-target data mapping, and the results of a sample transformation test. The full analysis is in the attached Excel workbook.

Source data: For this simulated project, the City of Melbourne open dataset "Street lights with emitted lux level (council-owned lights only)" (CC BY licence, 106,037 records) is used to represent the legacy asset export from the council's Access database.

1. **Profiling Summary**

Field	Finding	Impact
ext_id	100% populated, 106,037 unique values, 0 duplicates	Suitable as the asset identifier
prop_id, easting, northing	100% zero values	Not usable; excluded from migration
name, asset_clas, asset_type	100% blank	Asset class and type must be derived using default values
Geo Point	100% populated; latitude and longitude combined in one field	Must be split into two fields for GIS
Geo Point	8,632 records share coordinates with another record	Possible duplicates or multiple lights per pole; requires review
label (lux)	662 records show 0 lux	Possible faulty lights or data errors
xdate	Date stored as a number (YYYYMMDD); meaning not documented	Must be converted; meaning to be confirmed
(missing)	No suburb field in source data	Suburb reporting (REQ-015) depends on GIS suburb boundaries

2. **Key Findings**

- Only one source field (ext_id) is a reliable unique identifier.
- Location data exists only in Geo Point. Easting and northing are unusable.
- The source data does not support suburb-based reporting without GIS enrichment. This creates a dependency for REQ-015.

3. **Sample Transformation Test**

The mapping rules were tested on a sample of 50 records. All transformations produced the expected results: asset IDs were reformatted (e.g. 280 → SL-000280), coordinates were split into latitude and longitude, dates were converted (20140916 → 16/09/2014), and one 0-lux record was correctly flagged as "Inspect".

Observation: Some records have very low but non-zero lux values (e.g. 0.1). The threshold for flagging lights for inspection needs to be confirmed.

4. **Decisions Needed**

ID	Decision needed	Owner
DQ-01	Confirm default Asset_Class ("Lighting") and Asset_Type ("Streetlight")	Daniel Reid (Works Manager)
DQ-02	Confirm whether records sharing coordinates are duplicates or multiple lights per pole	Ben Okafor (GIS Officer)
DQ-03	Confirm the lux threshold for "Inspect" flag (0 lux or below 1 lux)	Daniel Reid (Works Manager)
DQ-04	Confirm the meaning of xdate (install date or survey date)	Ben Okafor (GIS Officer)
DQ-05	Provide GIS suburb boundary layer to derive suburbs	Ben Okafor (GIS Officer)

Related pages: 03 – RAID Log, 07 – Requirements Register

Supporting Workbook 📎

Northstar_Data_Profiling.xlsx

09 – Integration & Reporting Requirements

Project: Project Northstar | **Version:** 0.1 (Draft) | **Author:** Keerthi Ajantha Kuchipudi

Purpose: This page defines the interface requirements for the finance and GIS integrations, and the reporting requirements for the new asset and work order management system.

Related pages: 03 – RAID Log, 07 – Requirements Register, 08 – Data Profiling & Mapping

1. **Interface Overview**

Interface ID	Interface	Source → Target	Method	Frequency	Linked requirements
INT-01	Work order costs to finance	Asset System → Finance System	File-based batch (CSV)	Nightly, 11:00 pm	REQ-012, REQ-013, REQ-014
INT-02	Asset locations from GIS	GIS → Asset System	Map service (read-only)	Real-time when map is viewed	REQ-009, REQ-011

2. **INT-01: Work Order Costs to Finance**

Trigger: A work order is closed with at least one cost line (REQ-006).

Data sent (one row per cost line):

Field	Description	Example	Mandatory
Work_Order_No	Unique work order reference	WO-2026-00451	Yes
Asset_ID	Asset the work was done on	SL-000280	Yes
Cost_Code	Finance cost code for the work type	4120-LIGHT	Yes
Cost_Type	Labour or Material	Labour	Yes
Amount	Cost excluding GST (AUD)	185.50	Yes
Completion_Date	Date the work order was closed	2026-10-08	Yes

Supervisor_ID	Person who closed the work order	TWALSH	Yes

Business rules:

- Only work orders closed since the last successful batch are included.
- Each cost line is sent once only. Re-running a batch must not create duplicates (REQ-014).
- Costs must appear in the finance system by 7:00 am the next business day (REQ-012).

Error handling:

- Records that fail validation (e.g. missing or invalid cost code) are not imported. They are listed on a daily error report for the Finance Officer, with the reason for failure (REQ-013).
- Corrected records are included in the next nightly batch.
- If the whole batch fails, the system notifies the Finance Officer and the system administrator by email.

Open items:

- The finance system file format is to be confirmed by Sarah Chen (workshop action A-02).
- Vendor confirmation of batch capability is pending (workshop action A-04, RAID D-01).
- The cost code list for each work type is to be provided by Finance.

3. **INT-02: Asset Locations from GIS**

- The asset system displays assets on a map using the council's existing GIS layers (REQ-009).
- Asset coordinates are migrated as latitude and longitude, split from the legacy Geo Point field (see page 08).
- Suburbs will be derived from the GIS suburb boundary layer, as the legacy data has no suburb field (RAID I-02, DQ-05).
- The coordinate standard is to be confirmed by Ben Okafor (REQ-011, RAID R-02).

4. **Reporting Requirements**

Report ID	Report	Audience	Key fields	Filters	Frequency	Linked requirement

RPT-01	Open work orders dashboard	Works Manager, Supervisors	Work order no, asset ID, asset type, suburb, status, days open	Suburb, asset type, status	Real-time	REQ-015
RPT-02	Average response time	Works Manager	Month, number of work orders closed, average days from logged to closed	Month, asset type	Monthly	REQ-016
RPT-03	Maintenance costs by asset type	Works Manager, Finance Officer	Month, asset type, labour cost, material cost, total cost	Month, asset type	Monthly	REQ-017
RPT-04	Finance import error report	Finance Officer	Work order no, cost line, error reason, date	Date	Daily	REQ-013
RPT-05	Lights flagged for inspection	Supervisors	Asset ID, location, lux level, condition flag	Suburb	Weekly	Page 08 (0-lux finding, DQ-03)

Reporting dependencies:

- RPT-01 needs suburb data, which depends on the GIS suburb layer (RAID I-02).
- RPT-02 needs accurate "logged" and "closed" dates on every work order.
- RPT-03 depends on correct cost codes from Finance.

10 – Test Scenarios & Traceability

Project: Project Northstar | **Version:** 0.1 (Draft) | **Author:** Keerthi Ajantha Kuchipudi

Note: This is a simulated portfolio project. TC-10, TC-15 and TC-20 were executed for real as migration tests in Excel, using the City of Melbourne open dataset (see page 08). All other test results are simulated to show how testing and defects would be managed.

Purpose: This page lists the User Acceptance Testing (UAT) scenarios, defects raised, and a requirements traceability matrix showing that every requirement is covered by a test.

Related pages: 07 – Requirements Register, 08 – Data Profiling & Mapping, 09 – Integration & Reporting Requirements

1. **Test Scenarios**

Test ID	Requirement	Scenario	Steps	Expected result	Result
TC-01	REQ-001	Request cannot be saved without an asset	Log a new request and leave the asset field blank. Click Save.	Error message shown; request not saved	Pass (simulated)
TC-02	REQ-001	Request logged against a specific asset	Log a request against asset SL-000280. Click Save.	Request saved and linked to SL-000280	Pass (simulated)
TC-03	REQ-002	Unique reference number generated	Save two new requests.	Each request shows a different reference number	Pass (simulated)
TC-04	REQ-003	Search request by reference number	Search using the reference number from TC-03.	Request found and status shown (Open)	Pass (simulated)
TC-05	REQ-004	Assign work order to a crew	Open the work order and assign Crew A.	Crew and date recorded;	Pass (simulated)

				status changes to Assigned	
TC-06	REQ-005	Print job sheet with map	Print the job sheet for the assigned work order.	Job sheet shows asset ID, address, coordinates and map image	Fail – DEF-01 (simulated)
TC-07	REQ-006	Work order cannot close without costs	Try to close a work order with no cost lines.	System blocks closure and shows a message	Pass (simulated)
TC-08	REQ-009	Select asset on map	Open the map, click a streetlight.	Asset selected and its ID shown on the request	Pass (simulated)
TC-09	REQ-010	Asset type is mandatory	Create an asset without an asset type.	System blocks save	Pass (simulated)
TC-10	REQ-011	Coordinates split correctly for GIS	Apply the latitude/longitude transformation rule to legacy record ext_id 280.	Latitude -37.813836 and longitude 144.942292 in separate fields	Pass (executed in Excel, page 08)
TC-11	REQ-012	Costs reach finance by next morning	Close a work order with costs. Run nightly batch. Check finance system at 7:00 am.	Cost lines appear in finance with correct amounts	Pass (simulated)

TC-12	REQ-013	Failed records shown on error report	Close a work order with an invalid cost code. Run batch.	Record appears on error report with failure reason	Fail – DEF-02 (simulated)
TC-13	REQ-014	No duplicate costs on batch re-run	Re-run the same batch.	No duplicate cost entries in finance	Pass (simulated)
TC-14	REQ-015	Dashboard filters by suburb	Open dashboard and filter by suburb.	Only work orders in that suburb are shown	Blocked – RAID I-02
TC-15	REQ-018	Asset IDs are unique and complete	Count records and unique ext_id values in the legacy export.	106,037 records; 106,037 unique IDs; 0 duplicates	Pass (executed in Excel, page 08)
TC-16	REQ-018	Shared-coordinate records flagged, not merged	Check records that share coordinates (RAID R-04).	Records flagged for review; none merged automatically	Pass (simulated)
TC-17	REQ-019	Open requests migrated with status	Compare open requests in legacy data with new system.	All open requests present with correct status	Pass (simulated)
TC-18	REQ-021	Browser access without installation	Log in using Chrome and Edge on a council laptop.	Login successful; no installation needed	Pass (simulated)
TC-19	REQ-022	Customer Service cannot	Log in as Customer	Action not allowed	Fail – DEF-03 (simulated)

		close work orders	Service. Try to close a work order.		
TC-20	REQ-010	0-lux light flagged for inspection	Apply the condition flag rule to legacy record ext_id 11833 (0 lux).	Condition flag shows "Inspect" (SL-011833)	Pass (executed in Excel, page 08)

2. **Defect Log (simulated)**

Defect ID	Test ID	Description	Severity	Assigned to	Status
DEF-01	TC-06	Job sheet prints without the map image	Medium	Mark Ellis (Vendor)	Open
DEF-02	TC-12	Error report lists failed records but the failure reason is blank	High	Mark Ellis (Vendor)	Open
DEF-03	TC-19	Customer Service role can close work orders (role permission not configured)	High	Mark Ellis (Vendor)	Open

3. **Requirements Traceability Matrix**

Requirement	Priority	Source	Documented in	Test(s)	Status
REQ-001	Must	G-02, G-05	Page 07	TC-01, TC-02	Pass
REQ-002	Must	G-01	Page 07	TC-03	Pass
REQ-003	Must	G-01	Page 07	TC-04	Pass

REQ-004	Must	G-06	Page 07	TC-05	Pass
REQ-005	Must	G-02, D-01	Page 07	TC-06	Fail (DEF-01)
REQ-006	Must	G-03	Page 07	TC-07	Pass
REQ-007	Should	Workshop	Page 07	Not yet written (threshold TBC)	Not tested
REQ-008	Could	Workshop	Page 07	Not in Phase 1 test scope	Not tested
REQ-009	Must	G-02	Pages 07, 09	TC-08	Pass
REQ-010	Must	Workshop	Pages 07, 08	TC-09, TC-20	Pass
REQ-011	Must	R-02	Pages 07, 08, 09	TC-10	Pass
REQ-012	Must	G-03, D-02	Pages 07, 09	TC-11	Pass
REQ-013	Must	G-03	Pages 07, 09	TC-12	Fail (DEF-02)
REQ-014	Must	G-03	Pages 07, 09	TC-13	Pass
REQ-015	Must	G-04	Pages 07, 09	TC-14	Blocked (I-02)
REQ-016	Should	G-04	Pages 07, 09	Planned (UAT cycle 2)	Not tested
REQ-017	Should	G-04	Pages 07, 09	Planned (UAT cycle 2)	Not tested
REQ-018	Must	R-01	Pages 07, 08	TC-15, TC-16	Pass
REQ-019	Must	Workshop	Page 07	TC-17	Pass
REQ-020	Could	Workshop	Page 07	Not in Phase 1 test scope	Not tested
REQ-021	Must	Vendor	Page 07	TC-18	Pass
REQ-022	Must	Workshop	Page 07	TC-19	Fail (DEF-03)

REQ-023	Should	Brief	Page 07	Planned (performance testing)	Not tested

4. **Test Summary**

- **Must requirements:** 17 of 17 covered by at least one test.
- **Results for Must requirements:** 13 passed, 3 failed, 1 blocked.
- **Executed for real:** 3 migration tests (TC-10, TC-15, TC-20) were run in Excel on the open dataset, and all passed.
- **Should and Could requirements:** planned for UAT cycle 2 or out of Phase 1 scope.

Go-live readiness: Not ready. DEF-02 and DEF-03 (High severity) must be fixed and retested, and the GIS suburb layer (RAID I-02) must be available before REQ-015 can be tested.